

MGMT 67000 – Business Analytics

Syllabus: Fall 2025

Instructor

Prof. Will Wei Sun (Email: sun244@purdue.edu)

Lecture Time

MWF 9:50-11:20am EST

Classroom

RAWL 2070

Office

KRAN 359

Office hours

Tuesday 8:45am-9:45am or by appointment

Grader: Yu Gu (Email: gu342@purdue.edu)

INTRODUCTION

Data analysis and modeling are important skills for effective managerial decision making in business and industry. Advances in technology (computers, scanners, cell phones) have made significant amount of data available to managers. For example, the Dow Jones Industrial Average is one of the best-known and most widely watched indicators of the direction in which stock market values are heading. Administration and Congressional policymakers rely on statistics for budget decisions and related fiscal policy choices. The Federal Reserve System bases the monetary policy on data analysis. A manager needs to know if the manufacturing process is producing a quality product based on monitoring and assessing process performance. A sales manager has to develop tools to regularly monitor the performance of sales force. A manufacturer of certain electronic products needs to produce a forecast of future sales in order to decide whether or not to expand production. Banks use customer data to identify and design lucrative banking products. These are a few of the many examples from business where statistics can improve company performance.

The techniques learned in this course will help you infer data and as such make better informed decisions. The course covers basic probability, decision analysis, and statistical analysis (hypothesis testing and regression analysis). Probability models provide tools to handle uncertainty and risk. Statistical analysis focuses on the presentation of data and techniques to draw useful and valid inferences from data. Optimization models and decision analysis focus on techniques that use data to inform decision-making.

COURSE OBJECTIVES

The course emphasizes applications of data analysis through cases and computer exercises. The focus of the course is as much on modeling and presenting solutions to business problems as on understanding statistical methods. Areas covered by the course include descriptive statistics, exploratory data analysis, probability, decision analysis, estimation, hypothesis testing, and regression analysis. At the end of the course, you should have

- a good understanding of several commonly used statistical techniques in business;
- an ability to model and solve practical business problems using statistical methods;
- an ability to present your findings to an audience of managers/decision makers;
- a good understanding of the use of statistics software.

METHODS OF INSTRUCTION

The course is taught using a mix of lectures, case/data presentations, class discussions, interactive problem-solving, and computer exercises/demonstrations. Case/data set analyses provide opportunities for the creative application of statistics to unstructured management problems and also serve to illustrate the applications of statistics in managerial decision making. Homework exercises emphasize the understanding of concepts and aim to develop proficiency in translating business problems into statistical questions. Computer exercises familiarize the students with statistical software for data analysis.

Textbook and Course Materials for MGMT 67000

Statistics for Business and Economics, 14th edition by D. R. Anderson, D. J. Sweeney, T. A. Williams, Jeffrey D. Camm, and James J. Cochran, Cengage Publishing, ISBN-13 978-1337901062, 2020 (SBE)
<https://www.cengage.com/c/statistics-for-business-economics-14e-anderson/9781337901062/>

Software for MGMT 670

Minitab, Excel, Decision Tools (Excel add-ins PrecisionTree)

Minitab: MGMT 670 makes extensive use of Minitab for graphs and regression analysis. The software is accessible in the computer Lab in KRAN building and RAWLS Hall.

DecisionTools: MGMT 670 uses several Excel Addins such as PrecisionTree. This software is also available in the computer Lab in KRAN building and RAWLS Hall.

Virtual Access to Software: The Minitab softwares can be freely accessed via Purdue's goremote website, see page 5 of the syllabus. The Precision-Tree software in Palisade DecisionTools can also be freely accessed via Purdue network, see instruction here: <https://kcc.krannert.purdue.edu/Faq/Item/1017>. If you are off campus you will need to connect to Purdue VPN first.

GRADING POLICY/RULES

The following table lists the percentage of points allocated to each graded activity.

Class attendance	5%
Homework assignments (3 HWs)	30%
Midterm exam	20%
Final exam	45%

Attendance will be recorded **randomly** throughout the term. In the event of an illness, accident, or emergency, when circumstances permit, the student should show the evidence by emailing the instructor **before** the class starts. No make-up attendance will be given.

Late homework will NOT be accepted without prior permission.

You are discouraged from using AI tools (chat GPT and others) for homework, as it can hinder your ability to fully understand the material on your own.

Cellphones and internet access are prohibited during midterms and exams. However, you are allowed to use your own books/notes and personal calculators.

YOUR ROLE IN THE COURSE

Preparation Each student is expected to be prepared for each class, to contribute to class discussions, and to complete all assigned readings and exercises. Lecture material will be posted on the course website. Students are expected to be aware of the announcements, events, and files posted on the course website. The class will be divided into several teams. **Homework assignments will be handed in as a team.** Team homework assignments must be turned in on using Brightspace before their due time. The name of the submitted file should be **HW_X_Team_Y**, where X is the homework number, and Y is the team number.

Class and team-work is subject to the following rules:

- if a team member does not contribute, then his/her name should not appear on the assignments handed in, and he/she may do the entire assignment and hand it in separately;
- active participation in class discussions and teams is essential for clarifying difficult concepts. Thus, each team member will evaluate his or her team members at the end of the course.
- Cell phones may not be used during class. The use of laptops in class should be restricted to the course content in discussion. Disruptive use of personal property during class will be penalized by reducing the class participation grade.

COURSE HONOR POLICY

We expect and encourage students to discuss readings, case materials, and the concepts covered by the course with one another. However, do not falsely represent someone else's work as if it were your own. Students are expected to prepare homework, or other assignments without the assistance or reference to students who have taken the class before, prior semester's class notes and the like. A team is not allowed to cooperate with other teams. In case of violation, the instructor reserves the right to award zero points on the exam/assignment to the individual/team and refer the matter to the university for further action. Further, the use of Internet for finding solutions to cases and problems is prohibited. Specifically, students are expected to follow Purdue regulations governing student conduct (see <http://www.purdue.edu/univregs/studentconduct/index.html>). While working individually and with your team members:

Each team member is expected to understand, for each assignment, the reason for choosing a particular technique, the mechanics of the solution procedure, and the implication(s) of the final solution(s) and recommendation(s).

DEALING WITH CAMPUS EMERGENCY

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Brightspace and my email address are ways to get information about changes in this course.

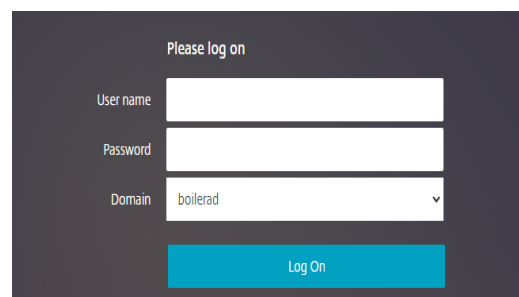
TENTATIVE COURSE SCHEDULE

Class	Date	Topic	Reading	Due
1	8/25	Syllabus Descriptive Statistics	SBE 3.1–3.5; M670_1(Descriptive Statistics).pdf	
2	8/27	Descriptive Statistics	SBE 3.1–3.5; M670_1(Descriptive Statistics).pdf	
3	8/29	Probability	SBE 4.1–4.5; M670_2(Probability).pdf	
4	9/3	Decision Analysis	SBE 19.1–19.4; M670_3(Decision Analysis).pdf	HW 1
5	9/5	Discrete Distributions	SBE 5.1–5.4; M670_4(Discrete Distributions).pdf	
6	9/8	Discrete Distributions	SBE 5.5; M670_4(Discrete Distributions).pdf	
7	9/10	Continuous Distributions	SBE 6.1 – 6.3, A 6.1, 6.2; M670_5(Continuous Distributions).pdf	
8	9/12	Continuous Distributions	SBE 6.1 – 6.3, A 6.1, 6.2; M670_5(Continuous Distributions).pdf	
9	9/15	Sampling Distributions and Estimation	SBE 7.1–7.5, M670_6(Estimation).pdf	HW 2
10	9/17	Sampling Distributions and Estimation	SBE 7.5–7.8; 8.1 – 8.4; M670_6(Estimation).pdf	
11	9/19	Midterm Review		
12	9/22	In-class Midterm		Midterm
13	9/24	Hypothesis Testing I	SBE 9.1–9.6; M670_7(Hypothesis I).pdf	
14	9/26	Hypothesis Testing I	SBE 9.1–9.6; M670_7(Hypothesis I).pdf	
15	9/29	Hypothesis Testing II	SBE 10.1–10.3; M670_8(Hypothesis II).pdf	
16	10/1	Simple Regression	SBE 14.1–14.4; M670_9(Simple Regression).pdf	
17	10/3	Simple Regression	SBE 14.5–14.9; M670_9(Simple Regression).pdf	
18	10/6	Multiple Regression	SBE 15.1–15.6; M670_10(Multiple Regression).pdf	HW 3
19	10/8	Multiple Regression	SBE 15.1–15.6; M670_10(Multiple Regression).pdf	
20	10/10	Final Review		
Final	TBD			Final Exam

Software Remote Instructions

Light Version

1. Visit <https://goremote.boilerad.purdue.edu/>
2. Fill in the correct information and select “Log On”
 - a. Username: **Career Account Username**
 - b. Password: **Career Account Password**
 - c. Domain: **boilerad**
3. Select “Use Light Version”
4. Select “Apps” at the top and then select the software you would like to use



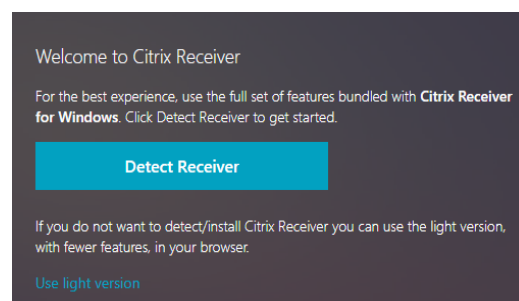
Please log on

User name

Password

Domain boilerad

Log On



Welcome to Citrix Receiver

For the best experience, use the full set of features bundled with **Citrix Receiver for Windows**. Click Detect Receiver to get started.

Detect Receiver

If you do not want to detect/install Citrix Receiver you can use the light version, with fewer features, in your browser.

[Use light version](#)

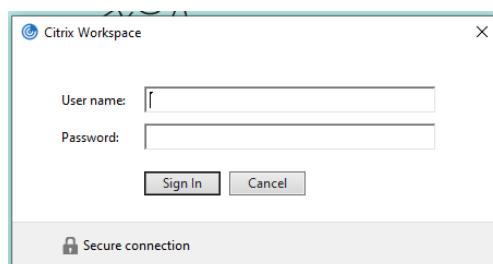
Citrix Workspace

1. Visit <https://www.citrix.com/downloads/workspace-app/> to download and install the **Citrix Workspace App** for the appropriate Operating System you are currently running. **NOTE:** ITaP lab machines already have this application installed
2. Run **Citrix Workspace**, and when the application comes up, enter “**goremote.itap.purdue.edu**” for the URL, select “Continue” and wait for the login prompt to enter your credentials
 - a. Username: **Career Account Username**
 - b. Password: **Career Account Password**
3. After the application finishes loading, select “Apps” at the top and then select the software you would like to use

Welcome to Citrix Workspace

goremote.itap.purdue.edu

Continue



Citrix Workspace

User name:

Password:

Sign In Cancel

Secure connection

Notice: VPN Connection required if off campus

Instructions for installing VPN found here:

<https://www.purdue.edu/advisors/documents/connect-virtual-private-network.pdf>